Arjuna JEE 2025

Maths

Basic Mathematics

DPP: 11

Q1 $\log_7 \log_7 \sqrt{7(\sqrt{7\sqrt{7}})} =$

(A) $3\log_2 7$

(B) $1-3\log_3 7$

(C) $1-3\log_7 2$

(D) $1-10\log_2 7$

Q2 Complete set of values of x for which $\log_x(x+3)$ is always defined is

(A) $x \in (-3,\infty)$

(B) $\mathrm{x} \in (0,\infty)$

(C) $\mathrm{x} \in (0,\infty)-\{1\}$

(D) $\mathrm{x} \in (-3,0)$

Q3 Complete set of values of x for which $\log_3(|x|-5)$ is defined is

(A) $\mathrm{x} \in (-\infty,-5] \cup [5,\infty)$

(B) $\mathrm{x} \in (5,\infty)$

(C) $x \in (-5,5)$

(D) $x \in (-\infty,-5) \cup (5,\infty)$

Q4 Value of: $\log\left(\frac{9}{20}\right)+\log\left(\frac{25}{18}\right)+\log\left(\frac{8}{5}\right) =$

(A) 0

(B) 1

(C) $\log 1800$

(D) 2

Q5 Simplified value of:

$\log_2 3 \cdot \log_3 5 \cdot \log_5 7 \cdot \log_7 8 =$

(A) 0

(B) 1

(C) 3

(D) Some irrational number

Q6 The number, $\mathrm{N}>0$

$N = \sqrt{(2+\sqrt{5})-\sqrt{(6-3\sqrt{5})+\sqrt{(14-6\sqrt{5})}}}$

Then, $\log_2 \mathrm{N}$ is

(A) 1 (B) 2

(C) 3 (D) 4

Q7 Product of solutions of equation $|\log_{10} 2\mathrm{x}| = 3$ is-

(A) 2 (B) $1/2$

(C) 4 (D) $1/4$

Q8 Sum of all the solution(s) of the equation $\log_{10}(\mathrm{x})+\log_{10}(x+2)-\log_{10}(5x+4)=0$ is-

(A) -1 (B) 3

(C) 4 (D) 5

Q9 If $\log_5(\log_5(\log_2 x))=0$, then the values of x, is

(A) 32 (B) 125

(C) 625 (D) 126

Q10 If $2^{\log_{10} 3\sqrt{3}} = 3^{\mathrm{k}^{\log_{10} 2}}$, then $\mathrm{k} =$

(A) $\frac{1}{2}$

(B) $\frac{3}{2}$

(C) 3

(D) 2

Q11 If $\log_x\left(\log_{18}(\sqrt{2}+\sqrt{8})\right)=\frac{1}{3}$, Then the value of $1000\mathrm{x}$ is equal to

(A) 8 (B) $1/8$

(C) $1/125$ (D) 125

Q12 Solve the equation for x :
$x^{\log x+4}=32$, where base of logarithm is 2 .

Q13 If $\log(x-y)-\log 5-\frac{1}{2}\log x-\frac{1}{2}\log y=0$
then

$$\frac{x}{y}+\frac{y}{x}=$$

(A) 25
(B) 26
(C) 27
(D) 28

Answer Key

Q1	(C)
Q2	(C)
Q3	(D)
Q4	(A)
Q5	(C)
Q6	(A)
Q7	(D)
Q8	(C)
Q9	(A)
Q10	(B)
Q11	(D)
Q12	1/32
Q13	(C)